FireScope

Interactive wildfire simulation and mitigation

CPU and Metal verified on Mac | CUDA validation pending

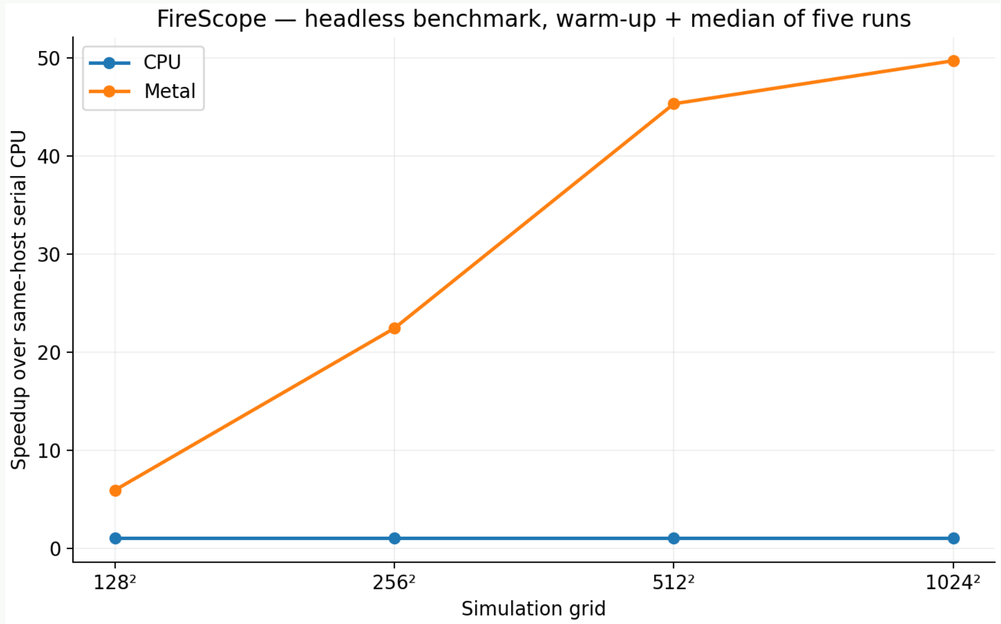


Branching forest, terrain shadows, fire and wind-driven smoke

45.31x

Metal step API speedup

512 by 512 cells, 500 updates



Apple M5 Pro | Median of five measured runs after one warm-up

Headless step batch time; rendering and final GPU readback excluded.

CPU counts are timed inside step; GPU counts occur during readback.

Problem and objective

Explain how fuel, wind, moisture and slope affect fire spread while comparing CPU and GPU computation. Support interactive, reproducible mitigation experiments.

Cellular model

Eight neighbors per cell. Ignition combines wind, slope, vegetation, density and moisture. Double buffers preserve synchronous updates; a seeded hash makes replay reproducible.

Architecture

Scenario and controls -> CPU / CUDA / Metal -> compact state -> OpenGL terrain, forest and particles.

Parallel computation

One GPU thread per cell with 16 by 16 groups. Nine Mac tests pass. CUDA and shared-memory optimization are awaiting NVIDIA validation; no CUDA speedup is claimed.

Mitigation comparison

A seeded 128 by 128 firebreak demo burned **11,985** cells versus **16,383** without intervention: **26.8% fewer**. Includes 728 cleared cells; 3,671 remain unburned.

SDGs and limitations

Educational alignment with SDG 13 Climate Action and SDG 15 Life on Land. Uncalibrated model with synthetic terrain; unsuitable for operational forecasts.

Model reference: Alexandridis et al. (2008), DOI 10.1016/j.amc.2008.06.046. Full sources and methods are in the project report.